

The Macro Alibi: Subjective Risk Attribution in Analyst Scenarios

Chen Wang and Kangying Zhou*

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Abstract

We study how analysts attribute subjective risk in scenario-based valuation reports. Using sell-side analysts' valuations and accompanying narratives, we document a Macro Bear Bias—a misattribution of the source of downside risk: analysts overweight macro risk in bear scenarios relative to base or bull scenarios. However, conditional on the same subsequent market state, realized CAPM R^2 is similar across realized bear, base, and bull outcomes, implying that analysts overstate the systematic component of downside risk. Consistent with this interpretation, the bear–base macro-attention gap positively predicts subsequent analyst forecast errors, with stronger predictive power in nonlinear specifications.

Keywords: macro bear bias, narrative finance, downside risk, analyst beliefs, large language model

JEL: C55, G11, G12, G14, G23

*Chen Wang is at University of Notre Dame, chen.wang@nd.edu. Kangying Zhou is at Texas A&M University, kangying.zhou@tamu.edu. We thank Jinchen Yang for excellent research assistance. We are grateful for helpful comments from Nick Barberis, Gerard Hoberg, Theis Jensen, Bryan Kelly, and Wei Wu.

1 Introduction

Understanding how market participants perceive risk is a central question in finance. Asset prices depend not only on fundamentals, but also on how investors and financial intermediaries interpret the source and meaning of uncertainty. Yet there is still limited evidence on subjective perceptions of risk in real-world market settings. A major reason is data limitation. Much of the existing evidence comes from surveys, which are informative but often coarse, infrequent, and detached from consequential market communication. Such data make it difficult to observe how agents explain risk, how they decompose uncertainty into different sources, and how those perceptions vary across scenarios.

Our paper brings to bear much richer information for studying risk perceptions by combining better data and better methods. On the data side, we use Morgan Stanley’s Risk-Reward Framework reports from 2007 to 2025, which require analysts to articulate bull, base, and bear scenarios together with narrative justifications. It makes analysts’ subjective views unusually transparent. Rather than observing only a point forecast, we observe how analysts describe upside and downside states, which risks they emphasize in each scenario, and how they verbally attribute the sources of potential outcomes. The structure of these reports therefore allows us to move beyond stated forecasts and examine the narrative architecture underlying perceived risk across scenarios. On the method side, we use large language models to systematically extract and classify the risks embedded in those narratives.

Together, these settings allow us to study a basic but underexplored question: how do analysts subjectively perceive and narrate risk across scenarios, and do those narratives accurately reflect the underlying sources of risk? More specifically, when analysts describe downside states, do they correctly condition on the market environment, or do they systematically rely on a macro-dominant narrative template that overstates the role of aggregate macroeconomic risk in bad outcomes? This question is important for at least three reasons. First, whether downside risk is perceived as macro or firm-specific directly affects valuation, discount rates, and expected returns. Second, analysts are influential information

intermediaries, so distortions in the way they frame risk can propagate into investor beliefs and market prices. Third, understanding whether downside narratives are well-calibrated helps distinguish rational state-contingent forecasting from narrative distortion and therefore sheds light on how beliefs are formed, communicated, and translated into forecast errors and asset-pricing outcomes.

Our central finding is that analysts perceive downside risk through a systematically distorted narrative lens: bear-case narratives place substantially greater emphasis on macroeconomic risks than base- or bull-case narratives. This asymmetry is not merely a difference in pessimistic tone. It reflects a distinct pattern of subjective risk attribution. To assess whether this pattern is justified, we compare ex ante narrative attribution with an ex post benchmark based on realized CAPM R^2 over the year following report issuance, conditional on the same market environment. If downside outcomes are truly more macro-driven, realized return variation should be more strongly explained by market movements in realized bear cases than in realized base or bull cases. Instead, we find that conditional on the same subsequent market state, realized CAPM R^2 is similar across realized bear, base, and bull outcomes, even though bear narratives remain persistently macro-heavy. We refer to this distortion as **Macro Bear Bias**: the tendency to over-attribute downside outcomes to aggregate macroeconomic risks in bear-case narratives.

This finding contributes to the literature in two ways. First, we show that understanding subjective risk perception requires studying narratives, not just forecasts. Numerical targets reveal what analysts expect, but narratives reveal how they map uncertainty into explanations. That mapping is economically meaningful because it determines whether downside risk is framed as macro, industry-specific, or firm-specific. Second, we show that the three-scenario structure of analyst reports provides a powerful way to identify miscalibration in subjective risk perception. By comparing bear, base, and bull narratives within the same report, we isolate the incremental macro attribution that analysts add when moving from the expected outcome to the downside. This form of narrative misattribution would be difficult to detect in single-scenario settings or with traditional survey data.

Scenario narratives help organize analysts’ broader valuation judgments, so an excessively macro-heavy bear narrative can spill over into the base-case forecast itself. Consistent with this mechanism, we show that the bear–base macro-attention gap predicts subsequent errors in analysts’ base-case forecasts. Firms whose downside narratives place unusually heavy emphasis on macro risk subsequently exhibit more positive forecast errors, indicating that analysts were too pessimistic *ex ante*. Moreover, this predictive relation is strongly nonlinear. Flexible nonlinear methods extract substantially more forecasting power than linear models, suggesting that the effect of Macro Bear Bias depends on how downside narratives interact with the prevailing market environment. In this way, the paper links subjective risk perception, narrative risk attribution, and economically meaningful forecast errors.

Two mechanisms may help explain Macro Bear Bias. One is cognitive: analysts may rely on the availability heuristic, with salient episodes such as the 2008 financial crisis and the 2020 COVID crash making macro-driven collapses especially easy to recall when constructing bear scenarios. The other is strategic: macro-based downside stories may be reputationally safer than sharp firm-specific negative claims, making them attractive under career concerns. Both mechanisms imply that analysts may place excessive macro emphasis on downside scenarios even when the actual driver of downside risk is not especially macro in nature.

More broadly, the paper studies whether one of the market’s key information producers correctly understands and communicates the driver of downside risk. That question matters for belief formation, price discovery, and the role of narratives in financial markets. By combining a uniquely informative setting with LLM-based measurement, we provide new evidence on how professional intermediaries perceive risk, how those perceptions become embedded in financial narratives, and how narrative distortions translate into forecast errors.

Literature

Our paper connects to four strands of literature. [Ang and Chen \(2002\)](#) document that correlations between U.S. equities and the market are substantially higher for downside

moves than for upside moves, and [Ang, Chen, and Xing \(2006\)](#) establish a downside beta premium in the cross-section of stock returns. [Lettau, Maggiori, and Weber \(2014\)](#) show that a conditional CAPM in which betas are elevated in bad states prices the cross-section of currency and equity returns, extending the partial rationality of state-dependent risk pricing to multiple asset classes. [Lewellen and Nagel \(2006\)](#) caution, however, that the conditional CAPM does not mechanically resolve standard asset-pricing anomalies, since beta variation would need to be implausibly large. [Lewellen, Nagel, and Shanken \(2010\)](#) further show that the standard cross-sectional testing approach is poorly suited to discriminating among conditional factor models. [Kacperczyk, Van Nieuwerburgh, and Veldkamp \(2016\)](#) and [Kacperczyk, Nieuwerburgh, and Veldkamp \(2014\)](#) show that fund managers optimally shift attention between aggregate market-timing and firm-specific stock-picking as a function of the business cycle: macro attention is concentrated in recessions, when systematic risk is most relevant, and dispersed in expansions. Our paper contrasts with this optimal attention-allocation result: analysts do not make the analogous shift in bear-case narrative construction. Bear narratives remain macro-heavy across all market regimes, suggesting that the narrative template is set at the time of scenario drafting without subsequent conditioning on realized states. These regularities constitute the genuine empirical foundation that analysts overgeneralize when constructing bear-case narratives. We complement this literature by characterizing the limits of that generalization: the empirical regularity in downside comovement is state-dependent, and analysts apply macro-heavy bear narratives unconditionally even when the realized evidence calls for state-dependent conditioning. The macro bear bias is not a departure from the Ang-Chen fact—it is the failure to condition on it.

A second strand studies the economic narratives analysts construct to justify their forecasts—the qualitative reasoning that accompanies and shapes quantitative outputs.¹ [Joos, Pi-](#)

¹A complementary and burgeoning literature studies subjective beliefs embedded in analyst *numerical* forecasts of earnings and long-term growth. [La Porta \(1996\)](#) shows that stocks with high analyst long-term growth forecasts earn negative abnormal returns, an early indication that analyst growth expectations are systematically optimistic. [Bordalo et al. \(2019\)](#) and [Bordalo et al. \(2024\)](#) develop the diagnostic expectations model and show it accounts for the overreaction pattern in analyst earnings forecasts and the associated stock

otroski, and Srinivasan (2016) show that analyst-constructed risk scenarios contain information about fundamental uncertainty not captured by standard quantitative forecasts. Ke (2024) use analyst narratives to recover subjective return beliefs, documenting that stated reasoning carries information beyond point estimates. Bradshaw et al. (2025) analyze qualitative signals in analyst reports at scale, showing that textual content unlocks dimensions of information not reflected in price targets. Bastianello, Décaire, and Guenzel (2024) show that forecasters with different internal models of the economy form persistently divergent expectations. We extend this literature by exploiting the *across-scenario* variation in narrative content—bear versus base versus bull—rather than the level of any single scenario. The three-scenario structure of the Morgan Stanley reports allows us to identify the incremental macro attribution that analysts add when moving from the expected outcome to the downside, which is precisely the miscalibration of interest and which a single-scenario analysis cannot detect.

Our paper also contributes to the emerging literature on narrative finance. Shiller (2019) argues that viral economic narratives—most potently, recession narratives—shape agents’ expectations and drive real outcomes. Our evidence complements this view: “recession risk” functions as a default bear-case template in analyst reports, applied regardless of whether the underlying downside risk is macro-driven. Addoum, Meursault, and Murfin (2024) document macro emphasis in debt investor narratives for corporate borrowers, providing complementary evidence from the bond side of the capital structure. We contrast with their level-based analysis by establishing that the equity bear narrative constitutes a state-dependent miscalibration: macro emphasis does not co-vary with the fraction of bear-case realizations that are genuinely systematic, which a level analysis cannot uncover.

Finally, our findings connect to the career concerns and herding literature in finance and economics. Holmström (1999) establishes the canonical model in which agents distort

return anomalies. Bouchaud et al. (2019) document that analysts are slow to revise earnings expectations in response to profitability news, generating the profitability anomaly. De la O and Myers (2021) use analyst forecasts to measure subjective cash flow and discount rate expectations, finding that cash flow news drives return variation more than standard rational-expectations models imply.

actions to signal ability, and [Scharfstein and Stein \(1990\)](#) show that career incentives lead agents to mimic each other even when private information would call for deviation. In the analyst context, [Hong and Kubik \(2003\)](#) show that analysts early in their careers issue more optimistic forecasts to manage the reputational cost of bold pessimistic calls, and [Chevalier and Ellison \(1999\)](#) document the same incentive structure for mutual fund managers. We reconcile this literature with the macro bear bias: attributing bear-case downside to macroeconomic forces shifts the perceived locus of the call from the analyst’s own judgment to an external macro shock, reducing reputational exposure if the forecast fails. This incentive to externalize blame complements the availability heuristic as a mechanism for why the macro bear bias is stable across firms, market regimes, and analyst cohorts.

The rest of the paper proceeds as follows. Section 2 documents the macro bear bias: we describe the data, construct the narrative attention measures, and show that bear-case macro attention is persistently elevated relative to the realized CAPM R^2 benchmark. Section 3 develops the conceptual framework, formalizes the bias, and presents empirical evidence on forecast error prediction. Section 4 discusses mechanisms. Section 5 concludes.

2 Documenting Macro Bear Bias

This section documents the paper’s main empirical result: Macro Bear Bias, the tendency for analysts to over-attribute downside outcomes to macroeconomic drivers in bear-case narratives. Analysts do not describe upside and downside risks symmetrically. Rather, bear scenarios receive disproportionately greater macro emphasis than base or bull scenarios. We begin by introducing the Morgan Stanley Risk-Reward reports, which provide an unusually rich setting for studying subjective risk attribution. We then use the narrative text in these reports to measure how analysts allocate attention across macro, industry, and firm-level drivers in bull, base, and bear cases. Finally, we compare these ex ante narrative attributions with an ex post realized benchmark based on CAPM R^2 , conditional on the same market environment. This comparison allows us to assess whether the macro-heavy framing of bear

scenarios reflects the true driver of downside outcomes or instead a systematic misattribution. The evidence points clearly to the latter: the persistent excess macro emphasis in bear narratives is the empirical signature of Macro Bear Bias.

2.1 Data

Our paper uses data from the Risk-Reward Framework in Morgan Stanley analyst reports. In total, we collected 63,191 Risk-Reward reports from January 2007 to May 2025.

Traditional sell-side research often relied on single-point price targets or coarse recommendations (e.g., “buy,” “hold,” or “sell”), which can convey a false sense of precision and provide limited information about the range of possible future outcomes. To address this limitation, Morgan Stanley introduced the Risk-Reward Framework in 2007, requiring analysts to present and justify 12-month stock-price expectations under three scenarios: bull, base, and bear. This framework was designed to provide a more informative and differentiated research product for institutional investors, who care not only about expected return but also about the distribution of potential outcomes and the sources of upside and downside risk. By explicitly presenting upside and downside scenarios, these reports help investors assess the skewness of expected payoffs, understand the assumptions underlying each case, and incorporate analyst views more directly into portfolio decisions. The reports are particularly valuable for our purposes because they provide both structured scenario-specific price targets and narrative discussion of the reasoning, key drivers, and investment debate behind those targets. For introductions to the framework, see [Weyns et al. \(2011\)](#) and [Srinivasan and Lane \(2011\)](#).

This dataset offers two main advantages. First, in addition to the base-case forecast, it contains analyst expectations under both bull and bear scenarios, which allows us to measure subjective upside and downside risk expectations separately. Second, the accompanying narratives explain the assumptions and reasoning behind each scenario. Using natural language processing (NLP) and large language models (LLMs), we can systematically extract and analyze these texts to identify the drivers of perceived upside and downside risks.

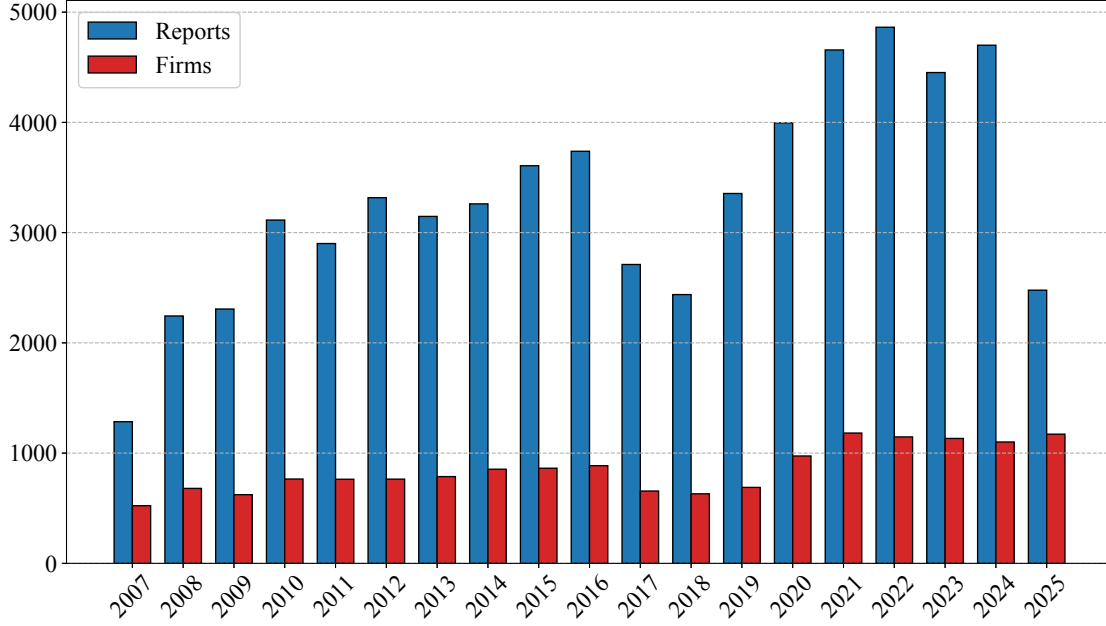


Figure 1 Histogram of Reports and Firms by Year.

Note: This figure shows the yearly count of Morgan Stanley Risk-Reward analyst reports and covered firms from January 2007 to May 2025.

To ensure the validity of the extracted scenario price targets, we retain only reports in which the base-case target lies between the bull-case and bear-case targets.² This filter yields a final sample of 62,571 reports. Table 1 reports the number of available Risk-Reward reports and the number of covered firms by year.

2.2 Textual Identification of Subjective Risks

We focus on the narrative text in each Morgan Stanley Risk-Reward report to understand how analysts justify their bull, base, and bear case expectations. Specifically, we examine whether analysts display a downside–upside asymmetry in the types of risks they emphasize across scenarios. To quantify the content of these narratives, we apply the GPT prompt described in Appendix Section A using the gpt-4o-mini model. Our main finding is that analysts place significantly greater emphasis on systematic downside risk in bear scenarios.

²To construct the dataset, we use three rule-based extraction procedures tailored to changes in Morgan Stanley report layouts over 2007–2014, 2015–2016, and 2017–2026. The procedures identify the relevant report section and extract the bull, base, and bear price targets. The resulting outputs are consolidated into a structured dataset for empirical analysis using Python packages including PyMuPDF and pdfplumber.

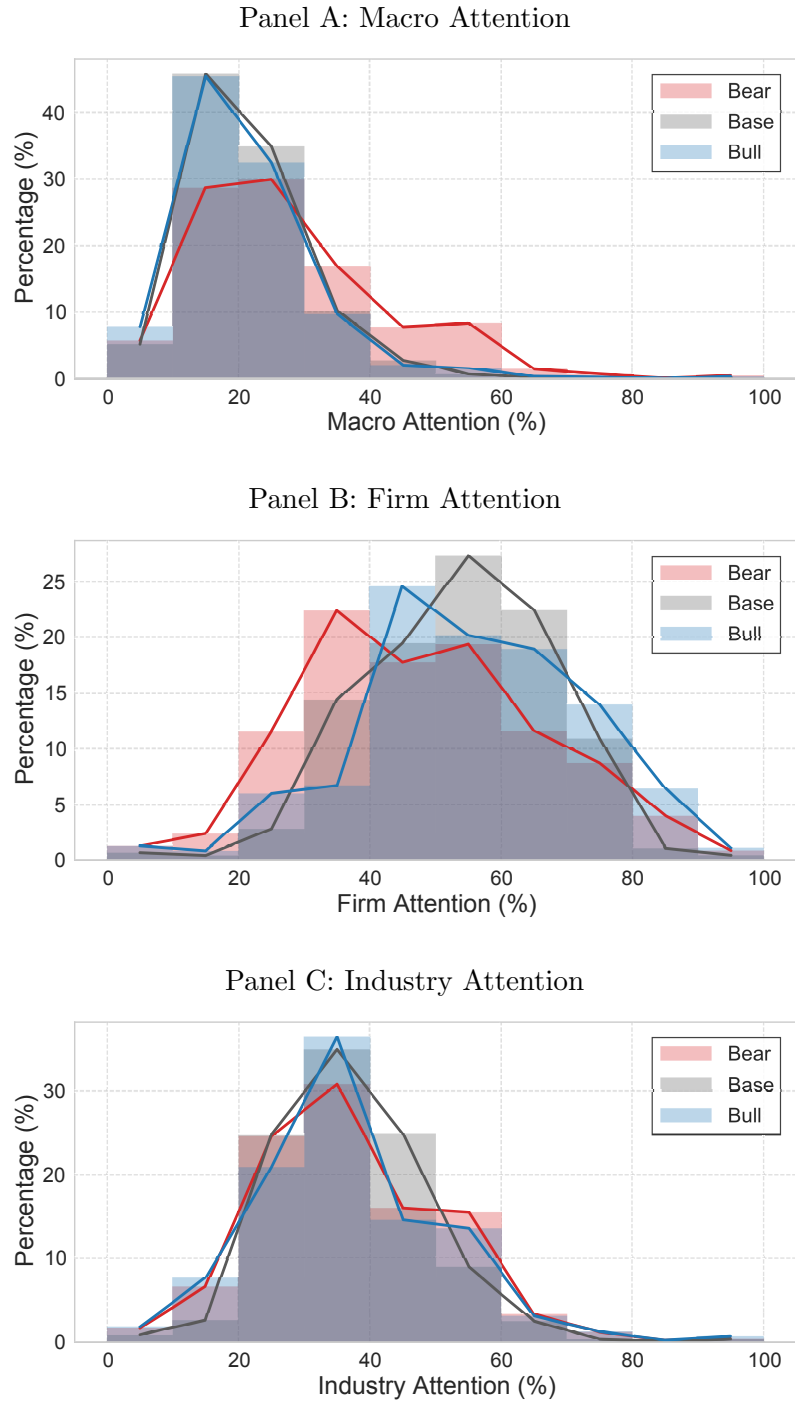


Figure 2 Attention Distribution to Macro, Firm, and Industry News.

Note: Report-level distribution of attention shares to firm-, industry-, and macro-level news across bull, base, and bear cases.

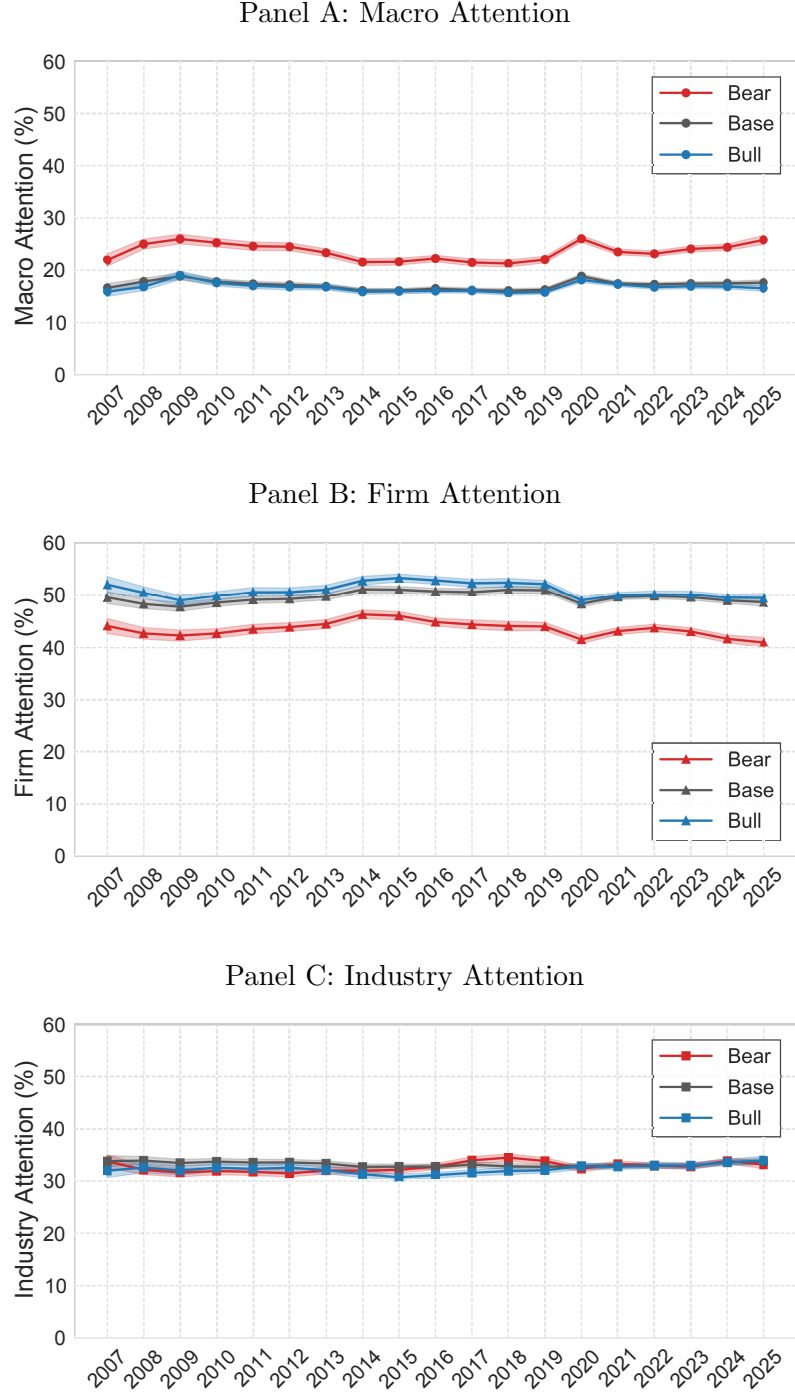


Figure 3 Time Series of Average Attention to Macro, Firm, and Industry News.

Note: Time series of report-level annual average attention shares to macro-, firm-, industry-level news. The shaded areas are 99% confidence interval of the average.

Figure 2 presents the report-level distributions of risk attention across scenarios. Panel A shows the distribution of attention to macro-level risks in bull, base, and bear cases. Panel B reports the corresponding distribution for firm-level risks, and Panel C does the same for industry-level risks. The figure reveals a clear pattern: relative to bull and base scenarios, bear scenarios assign substantially greater weight to macro-level risks and significantly less weight to firm-level risks. By contrast, the distributions of macro-level attention in bull and base scenarios are similar to each other, as are the distributions of firm-level attention across these two scenarios. Industry-level attention shows broadly similar distributions across all three scenarios. Figure 2 presents the report-level distributions of risk attention across scenarios. Panel A shows the distribution of attention to macro-level risks in bull, base, and bear cases. Panel B reports the corresponding distribution for firm-level risks, and Panel C does the same for industry-level risks. The figure reveals a clear pattern: relative to bull and base scenarios, bear scenarios assign substantially greater weight to macro-level risks and significantly less weight to firm-level risks. By contrast, the distributions of macro-level attention in bull and base scenarios are similar to each other, as are the distributions of firm-level attention across these two scenarios. Industry-level attention shows broadly similar distributions across all three scenarios.

Figure 3 complements this distribution evidence by showing the time-series evolution of average attention to macro-, firm-, and industry-level risks from 2007 to 2025. Panel A plots macro-level attention, Panel B plots firm-level attention, and Panel C plots industry-level attention. The time-series evidence reinforces the distributional results: throughout the sample period, analysts consistently devote more attention to macro-level risks in bear scenarios than in bull or base scenarios, while placing less attention on firm-level risks in bear scenarios. In contrast, attention patterns in bull and base scenarios remain broadly similar over time for both macro- and firm-level risks. Industry-level attention again appears relatively stable and similar across bull, base, and bear cases. For completeness, Figure A.1 reports the distribution at each point in time.

Table 1 Attention Across Scenarios

Note: This table reports panel regressions of attention shares on scenario indicators. The dependent variables are macro attention (cols 1–2), firm attention (cols 3–4), and industry attention (cols 5–6). Base scenario is the omitted category; the constant reports the mean attention share for the Base scenario. Bear and Bull coefficients measure the difference relative to Base. All specifications include firm fixed effects; even columns add year-month fixed effects. Standard errors are clustered by firm and month. *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
	Macro	Macro	Firm	Firm	Industry	Industry
Bear	6.4660*** (34.04)	6.4660*** (34.04)	-6.0675*** (-31.52)	-6.0675*** (-31.52)	-0.3996** (-2.53)	-0.3996** (-2.53)
Bull	-0.3961*** (-5.50)	-0.3961*** (-5.50)	1.1547*** (10.14)	1.1547*** (10.14)	-0.7614*** (-7.17)	-0.7614*** (-7.17)
Constant	17.1827*** (212.61)	17.1827*** (260.51)	49.6471*** (523.20)	49.6471*** (674.05)	33.1652*** (485.62)	33.1652*** (544.51)
Firm FE	Y	Y	Y	Y	Y	Y
Month FE		Y		Y		Y
Observations	187,713	187,713	187,713	187,713	187,713	187,713
R^2	0.2766	0.2830	0.2393	0.2465	0.1440	0.1480

2.3 Realized CAPM R^2

To assess whether analysts are correct to place greater weight on macro risks in bear scenarios than in base or bull scenarios, we construct an *ex post* realized benchmark based on the realized CAPM R^2 in each case. The intuition is straightforward: if macro risks truly matter more in firms' downside states, then a firm's realized return variation should be more strongly explained by market-wide movements when the report ultimately realizes a bear outcome than when it realizes a base or bull outcome. Accordingly, we test whether the realized CAPM R^2 is significantly higher for realized-bear reports than for realized-base or realized-bull reports. A higher realized R^2 in bear outcomes would support analysts'

emphasis on macro risks in downside scenarios; otherwise, it would suggest that analysts overweight macro risks in bear-case narratives.

Our empirical procedure has three steps. First, for each report, we compute the realized ex post CAPM R^2 by regressing the firm’s daily excess returns on daily market excess returns over the 252 trading days following the report release. This R^2 measures the extent to which the firm’s subsequent stock return variation is explained by systematic market movements.

Second, we classify each report as realized-bear, realized-base, or realized-bull according to the relationship between the analyst’s target prices and the stock’s realized price one year after the report release. A report is classified as realized-bear if the realized price is less than or equal to the bear-case target price. It is classified as realized-bull if the realized price is greater than or equal to the bull-case target price. If the realized price lies strictly between the bear and bull targets, the report is classified as realized-base.

Third, we aggregate these report-level realized R^2 measures to the firm level. For each firm, the realized-bear R^2 is the average realized CAPM R^2 across all of its realized-bear reports. Likewise, the realized-base and realized-bull R^2 are defined as the average realized CAPM R^2 across the firm’s realized-base and realized-bull reports, respectively. Constructing the measures at the firm level allows us to compare whether the realized R^2 is higher in bear outcomes than in base or bull outcomes for the same firm, thereby reducing concerns that cross-firm differences drive the results.

To make these comparisons within a common market environment, we further construct firm-level realized R^2 measures separately by market condition. Specifically, under a given market state, we restrict the sample to firms that experience all three realized report types—bear, base, and bull—within that same market condition. For example, when analyzing periods in which the market excess return over the 252 trading days after the report release lies between -5% and 5% , we include only firms that have realized-bear, realized-base, and realized-bull reports within that market-return range. This restriction ensures that comparisons across realized cases are not mechanically driven by differences in overall market environments.

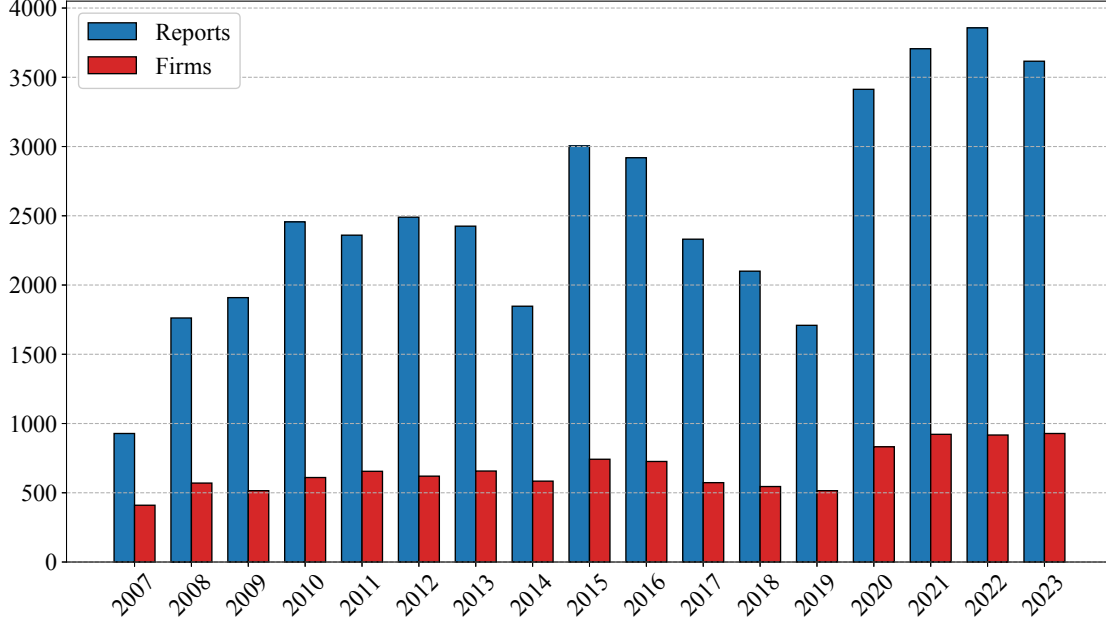


Figure 4 Yearly Distribution of Reports and Firms after Matching with Returns.

Note: This figure shows the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms after matching the reports with one-year-ahead firm returns.

We obtain stock return data from the CRSP database. After matching Morgan Stanley reports to 252-trading-day forward returns following the report release date, the sample contains 45,348 reports in total. Figure 4 presents the annual number of Morgan Stanley Risk-Reward analyst reports and the number of covered firms in the matched sample.

To mitigate distortions from stock splits and reverse splits, we construct an alternative measure of the one-year-ahead realized price by compounding daily returns excluding dividends, denoted by $r_{t,x}$, over the 252 trading days after the report release:

$$\text{constructed realized price} = \text{current price} \times \prod_{t=1}^{252} (1 + r_{t,x}).$$

If the realized price in one year differs from the constructed realized price by more than 10% in either direction, we replace the observed realized price with the constructed realized price. This adjustment helps correct cases in which stock splits or reverse splits mechanically distort the reported price level.

Table 2 Summary Statistics

Note: This table reports summary statistics for the Morgan Stanley Risk-Reward report sample from 2007 to 2025. Panel A reports report-level attention shares (percentages) by scenario. Panel B reports firm characteristics at the report date. All attention shares are extracted by GPT using the prompt in Appendix A.

Variable	N	Mean	Median	Std	Min	P25	P75	Max
Panel A: Narrative Attention								
Bull: Macro	45348	16.70	10.00	11.11	0.00	10.00	20.00	100.00
Bull: Industry	45348	32.28	30.00	14.32	0.00	20.00	40.00	100.00
Bull: Firm	45348	51.01	50.00	17.31	0.00	40.00	60.00	100.00
Base: Macro	45348	17.02	15.00	9.46	0.00	10.00	20.00	100.00
Base: Industry	45348	33.13	30.00	11.47	0.00	25.00	40.00	100.00
Base: Firm	45348	49.85	50.00	14.41	0.00	40.00	60.00	100.00
Bear: Macro	45348	23.35	20.00	15.11	0.00	10.00	30.00	100.00
Bear: Industry	45348	32.62	30.00	14.04	0.00	20.00	40.00	100.00
Bear: Firm	45348	44.03	40.00	18.32	0.00	30.00	60.00	100.00
Panel B: Returns and Forecast Variables								
Return	45348	0.12	0.07	0.74	-3.29	-0.17	0.31	74.14
Subjective Return	45348	0.18	0.10	1.43	-4.67	-0.01	0.22	97.63
Forecast Error	45348	-0.06	-0.02	1.59	-96.60	-0.30	0.26	74.03
$\hat{\beta}_{\text{CAPM}}$	42626	1.13	1.08	0.44	-1.04	0.83	1.36	5.29
R^2_{CAPM}	42626	0.32	0.30	0.18	0.00	0.18	0.45	0.86

Table 2 reports summary statistics of core narrative and firm-level data. In Panel A highlights the narrative asymmetry across scenarios. Bear-case macro attention averages 23.4%, compared with 17.0% and 16.7% in the base and bull cases, respectively—a bear–base gap of approximately 6.3 percentage points. The mirror image appears in firm-level attention: bear-case firm attention averages 44.0%, roughly 6 percentage points below the base-case mean of 49.9%, while industry attention is nearly identical across all three scenarios. Panel B shows that analyst base-case targets are slightly optimistic on average (mean forecast error of -0.06), and that the coverage universe skews toward high-beta firms (mean $\hat{\beta}_{\text{CAPM}} = 1.13$), consistent with Morgan Stanley’s large-capitalization focus.

Figure 5 illustrates how realized bear, base, and bull reports are distributed across

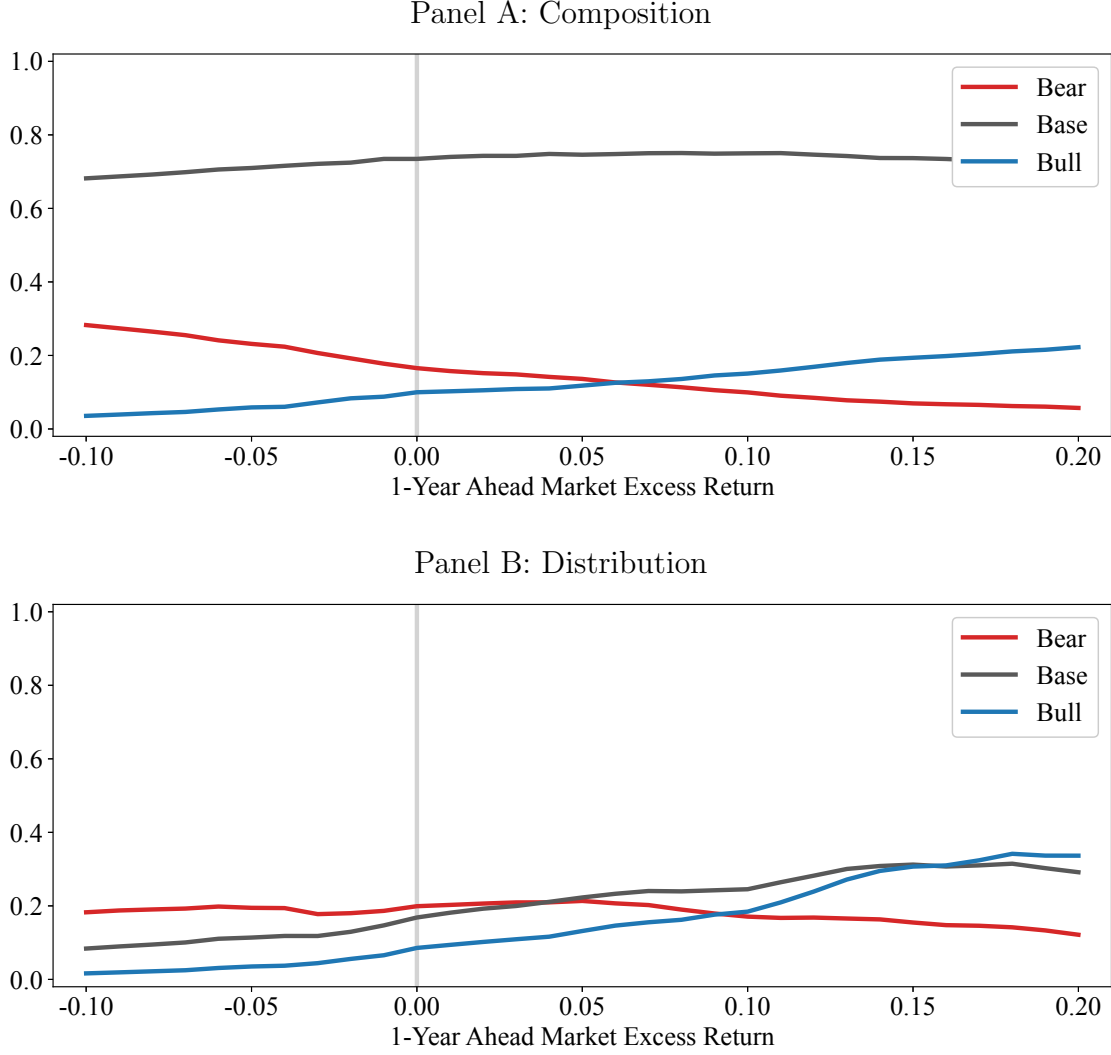


Figure 5 Composition and Distribution of Realized Bear, Base, and Bull Reports Across Market States.

Note: This figure plots the composition and distribution of realized bear, base, and bull reports across market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. Panel A shows the composition of bear, base, and bull reports within each return state:

$$\frac{\text{Number of realized bear/base/bull reports in a return state}}{\text{Total number of reports in that market excess return}}.$$

Panel B shows the distribution for realized bear, base, and bull reports:

$$\frac{\text{Number of realized bear/base/bull reports in a market excess return state}}{\text{Total number of realized bear/base/bull reports}}.$$

different market states. To characterize market conditions, we group reports according to the forward 252-day market excess return following the analyst report release. For each point on the horizontal axis, we consider reports whose forward market excess returns fall within a centered 10% window around that point.

The two panels present the same data from complementary perspectives. Panel A focuses on the composition of realized outcomes within each market state. For a given market-return state, it reports the shares of realized bear, base, and bull reports among all reports that fall into that state. This perspective shows how the relative likelihood of each realized outcome changes as aggregate market conditions become weaker or stronger. The figure shows a clear pattern: as future market excess returns rise, the share of realized bear reports declines, while the share of realized bull reports increases. In contrast, the share of realized base reports remains comparatively stable across market states. This pattern is intuitive. When the overall market performs poorly, more stocks are likely to end up near analysts' downside scenarios, making realized bear outcomes more common. When the market performs strongly, more stocks are likely to reach their upside scenarios, increasing the frequency of realized bull outcomes.

Panel B instead focuses on how each realized outcome is distributed across market states. For realized bear, base, and bull reports separately, it plots the fraction of that report type that falls into each market-return state. This perspective highlights the market environments in which each realized outcome is most likely to occur. Realized bear reports are spread more broadly across market states, resulting in a relatively flat profile. By contrast, realized base reports, and especially realized bull reports, become more concentrated in stronger market states. This pattern suggests that the distribution of forward market excess returns is tilted toward positive values, so realized base and bull outcomes are more likely to occur when subsequent market conditions are favorable.

Taken together, the two panels provide a more complete picture of the relation between realized report outcomes and subsequent market conditions. The figure shows that realized outcomes are systematically linked to the broader market environment: realized bear

outcomes are more prevalent when subsequent market conditions are weak, while realized bull outcomes become both more common and more concentrated when subsequent market conditions are strong. This implies that, when comparing realized bear, base, and bull benchmarks, it is important to condition on market state; otherwise, observed differences may conflate firm-level realization with variation in aggregate market performance.

2.4 Empirical Evidence on Macro Bear Bias

In this section, we examine whether analysts are justified in assigning substantially greater attention to macro risks in bear cases than in base or bull cases. We show that they are not. Instead, analysts systematically overweight macro risk in bear scenarios. We refer to this distortion as Macro Bear Bias: the tendency to over-attribute downside outcomes to aggregate macro risks in bear-case narratives.

To ensure an apples-to-apples comparison with realized R^2 , we construct firm-level ex ante macro attention for each case (bear, base, and bull) as the average macro attention across all reports of the firm that fall within the same market state. Thus, the comparison with the ex post realized R^2 is based on the exact same subset of reports. Figure 6 plots realized R^2 and macro attention conditional on market state. For each point x on the horizontal axis, we consider reports whose forward 252-day market excess returns fall within a centered 10% window, $[x - 5\%, x + 5\%]$, around that point.

Panel A shows the average CAPM R^2 over the 252 trading days after report release for realized bear (red), realized base (yellow), and realized bull (green) cases. The figure shows that, conditional on the same market state, the three realized R^2 series are very similar. At the same time, all three decline as subsequent market excess returns become weaker. This pattern indicates that macro risk exposure is driven primarily by the realized market environment, rather than by whether the report is ultimately classified as realized bear, base, or bull. Panel B shows the average ex ante macro attention that analysts assign to the bear, base, and bull cases.

Two findings stand out. First, the three attention series are relatively flat across market

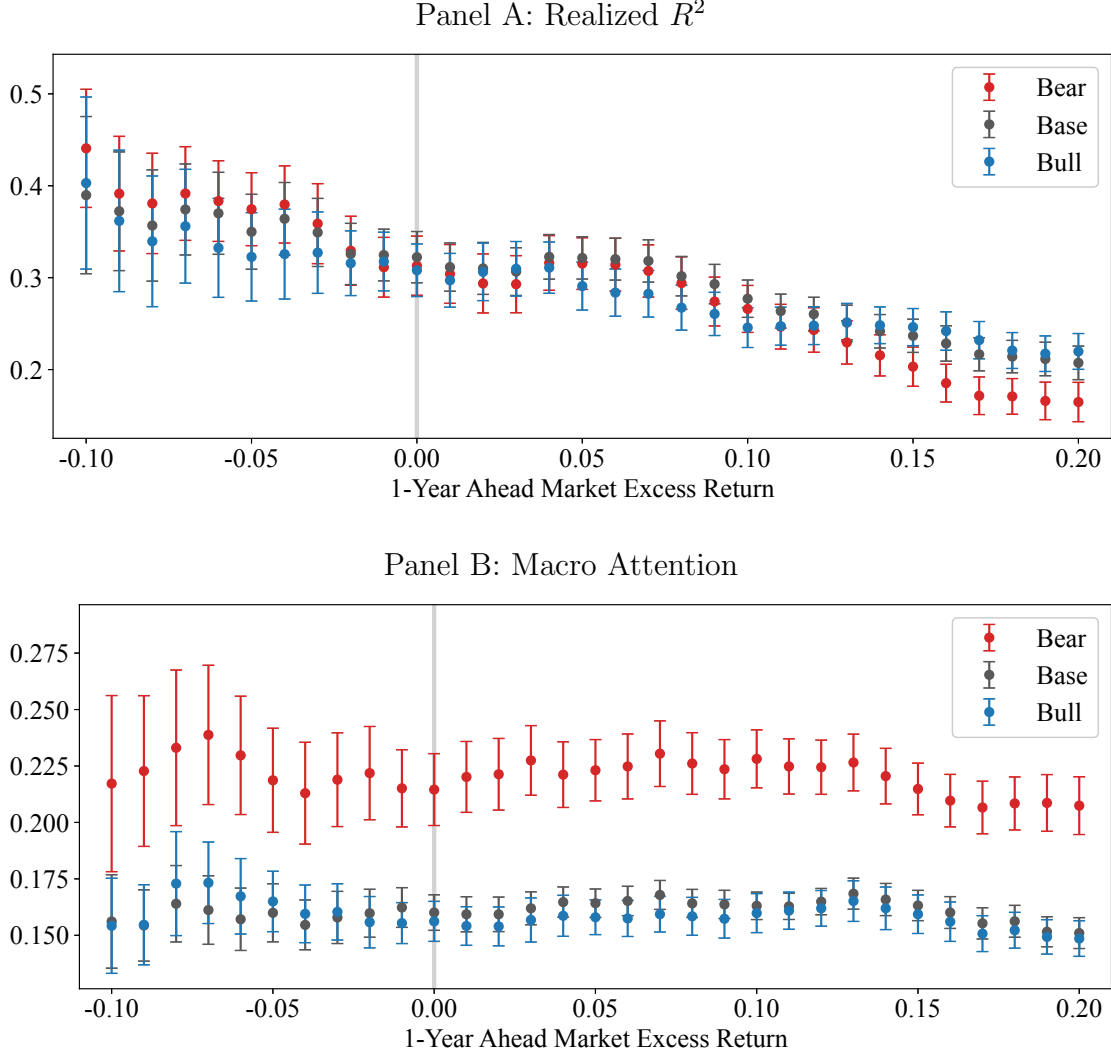


Figure 6 Realized R^2 and Macro Attention Conditional on Market States.

Note: This figure shows the realized R^2 and macro attention conditional on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention within each realized scenario (bear, base, and bull). Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear/base/bull cases. Error bars show 95% confidence intervals from constant-only regressions.

states compared with their realized R^2 benchmarks. This suggests that analysts do not sufficiently adjust their attribution of macro risk to the realized market environment. Second, macro attention in the bear case is consistently and substantially higher than in the base or bull case. This persistent wedge is the empirical signature of Macro Bear Bias: even when

the realized macro component of risk is similar across scenarios conditional on market state, analysts continue to describe downside outcomes as disproportionately macro-driven.

Figure 7 makes this pattern especially clear by plotting the bear-base gap in macro attention together with the corresponding bear-base gap in realized R^2 across market states. The realized R^2 gap is close to zero when the subsequent annual market excess return is below 12%, and turns negative when the market excess return is above 12%. In contrast, the bear-base gap in macro attention remains significantly positive throughout. In other words, analysts consistently portray the bear case as much more macro-driven than the base case, even though the realized difference in macro risk is economically small or even negative, depending on the market state. This divergence provides direct evidence of Macro Bear Bias.

These patterns point to two possible explanations for Macro Bear Bias. The first is that analysts implicitly treat realized macro risk as roughly constant across market states, while at the same time mechanically assigning extra weight to macro risk in bear cases. Under this explanation, analysts do condition on downside scenarios, but they fail to recognize that the importance of macro risk varies with the market environment. The second possibility is that analysts understand that realized R^2 declines with market returns, but still attach a persistently high probability to a severely negative future market outcome regardless of the prevailing market state. Under this interpretation, the elevated macro attention in bear cases would reflect a belief that downside scenarios are tightly linked to large aggregate market declines.

The evidence is more consistent with the first explanation. If the second explanation were correct, two implications should follow. First, macro attention in the base case should move with market state and roughly track the realized base-case R^2 , which declines as subsequent market conditions improve. Second, macro attention in the bull case should be materially lower than in the base case, since that explanation implies analysts understand that realized R^2 falls with market returns and should therefore assign less macro risk to more favorable scenarios. The data support neither implication. Macro attention in the base case remains largely flat across market states, rather than tracking the realized base-case R^2 . In addition,

conditional on the same market state, the realized R^2 series for the base and bull cases are similar, which is difficult to reconcile with the assumption that analysts meaningfully incorporate the relation between realized R^2 and market returns. Taken together, these findings suggest that analysts are not merely mapping different scenarios into different expected market states; rather, Macro Bear Bias reflects a systematic tendency to over-attribute downside outcomes to macro risks in bear-case narratives.

One interpretation of the first explanation is that the bias is state-independent, which points to two cognitive mechanisms. The first is the availability heuristic (Tversky and Kahneman, 1973). When analysts construct a bear case, they may rely on resemblance to a canonical macro-crisis prototype—a severe, overwhelmingly systematic downturn—rather than on a state-contingent assessment of systematic risk. Because prototype matching is unconditional, it naturally predicts the regime-independent pattern we document. The contrast with idiosyncratic framing follows directly: firm-specific disasters do not share a common prototype and therefore are less likely to shape bear-case narratives. The second mechanism is anchoring with insufficient adjustment (Tversky and Kahneman, 1974): even when firm-specific information calls for a different framing, analysts may adjust too little away from the initial macro template.

3 Conceptual Framework and Forecast Error Evidence

We now formalize the bias documented in Section 2. Analysts routinely describe bear, base, and bull cases to explain the range of possible future outcomes. In principle, these narratives should reflect how different sources of risk contribute to firm value conditional on the prevailing market environment. In practice, however, analysts appear to assign disproportionately high macro emphasis to downside scenarios, even when the conditional macro component of risk is not especially large.

A central implication of this framework is that the bias does not remain confined to the bear narrative itself. Because scenario narratives shape analysts’ overall valuation judgments,

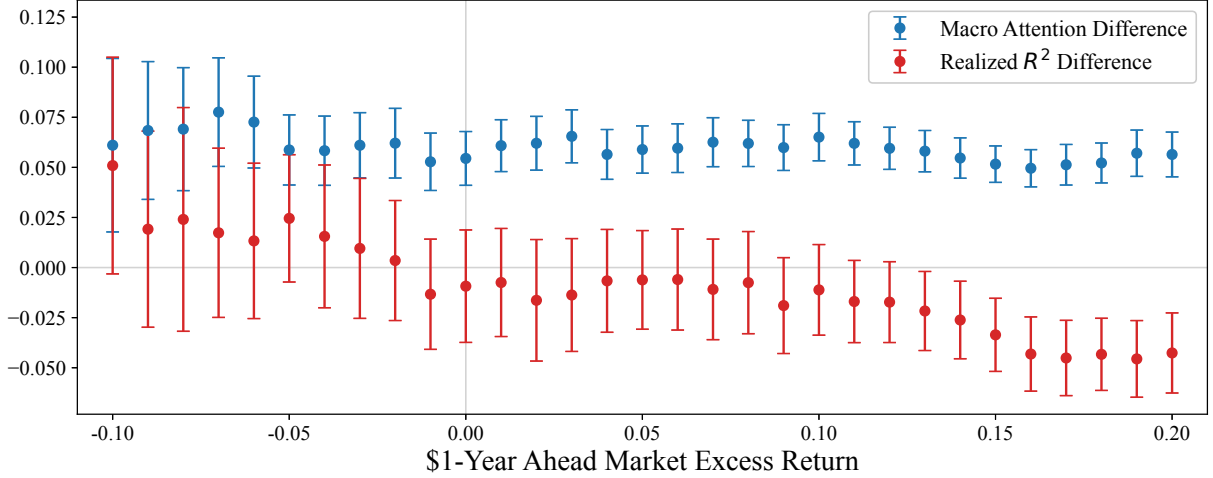


Figure 7 Bear – Base Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bear–base differences in macro attention and realized CAPM R^2 on different market states. For each point x on the horizontal axis, we consider the reports within 10% bins ($[x - 5\%, x + 5\%]$) of forward 252-day market excess returns following the analyst report release. We apply the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bear and base cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bear or realized base cases. The plotted values represent the bear-minus-base differences. Error bars show 95% confidence intervals from constant-only regressions.

excessive macro emphasis in the bear case can spill over into the analyst’s base-case forecast, making it too pessimistic relative to subsequently realized outcomes. This link connects Macro Bear Bias to the prediction of analyst forecast errors.

3.1 Benchmark: Conditional Attribution of Risk

Consider firm i ’s future return:

$$R_{i,t+1}^{\text{realized}} = \alpha_{i,t} + \beta_i M_{t+1} + \varepsilon_{i,t+1}, \quad (1)$$

where M_{t+1} denotes the future market excess return, β_i measures the firm’s exposure to aggregate macro conditions, and $\varepsilon_{i,t+1}$ captures firm-specific shocks. The key feature of the benchmark is that the importance of macro risk is conditional: it should be evaluated relative to the market state prevailing when the analyst forms the forecast.

Let s_t denote the information available at time t about the market environment. In a rational benchmark, the level of macro attention that an analyst assigns to scenario $k \in \{\text{bear, base, bull}\}$ should depend on the expected conditional importance of aggregate market forces for that scenario. Formally, let $\text{Macro Importance}_{i,t+1} \equiv \beta_i^2 \text{Var}(M_{t+1}) / \text{Var}(R_{i,t+1})$ denote the fraction of firm-level return variance attributable to market movements. The rational benchmark is:

$$A_{i,k,t} = g(\mathbb{E}_t[\text{Macro Importance}_{i,t+1} \mid s_t, k]), \quad (2)$$

where $g(\cdot)$ is an increasing mapping from the underlying importance of macro risk into narrative emphasis. Under this benchmark, two properties should hold.

First, state dependence: macro attention should vary with the prevailing market environment, because the contribution of macro forces to firm outcomes is not constant across states. Second, scenario discipline: differences in macro attention across bear, base, and bull cases should reflect genuine differences in the conditional relevance of macro risk, rather than a mechanical tendency to describe bad outcomes as macro-driven. In other words, the bear case may legitimately receive more macro emphasis in some settings, but only to the extent warranted by conditional macro exposure.

This benchmark provides the reference point for interpreting the data. If analysts are well calibrated, ex ante macro attention should track the ex post macro-related component of realized variation conditional on market state. If it does not, that gap reflects misattribution.

3.2 Macro Bear Bias: Over-attributing Macro Risks in Bear Narratives

Macro Bear Bias arises when analysts depart from this benchmark by assigning excessive macro weight specifically to the bear narrative. Formally, let observed macro attention satisfy

$$A_{i,k,t} = A_{i,k,t}^* + b_i \cdot \mathbf{1}\{k = \text{bear}\}, \quad b_i > 0. \quad (3)$$

The parameter b_i captures the analyst’s excess tendency to explain downside scenarios using aggregate macro forces, beyond what is justified by the conditional importance of macro risk.

Under this formulation, the distortion is not that analysts discuss macro risk per se. Rather, it is that they over-explain downside outcomes through macro forces. The bear narrative becomes disproportionately macro-heavy even when, conditional on the same market state, the realized macro component of outcomes is similar to that in the base or bull case. This is why Macro Bear Bias is best understood as a misattribution bias.

The empirical counterpart of this bias is the bear–base macro-attention gap:

$$\text{MacroGap}_{i,t} = \text{MacroAttention}_{i,t}^{\text{bear}} - \text{MacroAttention}_{i,t}^{\text{base}}. \quad (4)$$

A larger $\text{MacroGap}_{i,t}$ indicates that the analyst places substantially greater macro emphasis on the downside scenario than on the central scenario. Under the benchmark, this gap should reflect genuine conditional differences in macro relevance. Under Macro Bear Bias, however, the gap is too large because the bear narrative receives an additional, bias-driven macro overlay.

3.3 Economic Intuition

The framework distinguishes between two conceptually separate objects. The first is the true conditional importance of macro risk for firm outcomes. The second is the analyst’s narrative attribution of macro risk across scenarios. If analysts are well calibrated, these two objects should move together. If macro forces are genuinely more important in a given state, analysts should emphasize them more. If they are not, macro emphasis should be lower.

Macro Bear Bias breaks this alignment. Analysts appear to apply a coarser attribution rule: downside scenarios are treated as especially macro-driven almost by construction. As a result, narrative macro attribution becomes too insensitive to the actual market state and too concentrated in bear cases. The problem is not simply that analysts mismeasure macro

risk on average; the problem is that they do so asymmetrically, by disproportionately loading macro explanations into downside narratives.

This asymmetry matters economically because scenario narratives are not just descriptive. They help organize analysts’ thinking about valuation. A bear case that is described as strongly macro-driven can influence how the analyst interprets the central tendency of future outcomes. In that sense, Macro Bear Bias may contaminate not only the tail narrative but also the base-case forecast itself. The analyst’s central forecast then inherits some of the excessive macro pessimism embedded in the bear narrative. Two channels provide microfoundations for this spillover.

The primary channel operates through discount rates. A bear case described as predominantly macro-driven leads the analyst to perceive elevated systematic risk for the firm. Because the “likely” downside is framed as correlated with the market, the analyst implicitly adopts a higher equity risk premium when discounting base-case cash flows. To see this concretely, let the base-case price target be $P^{\text{base}} = \sum_{\tau} CF_{\tau}/(1 + r_{\tau})^{\tau}$, where $r_{\tau} = r_f + \tilde{\beta}_i \cdot \text{ERP}$. Macro Bear Bias inflates the analyst’s perceived beta:

$$\tilde{\beta}_i = \beta_i + \phi \cdot \text{MacroGap}_{i,t}, \quad \phi > 0. \quad (5)$$

The higher perceived beta raises the discount rate and depresses P^{base} even if point estimates of base-case cash flows are unchanged. Crucially, this channel scales with the firm’s true systematic exposure β_i : the discount-rate wedge is larger for high-beta firms, generating the $\beta_i \times \text{MacroGap}_{i,t}$ interaction tested in the empirical specifications below. This microfoundation maps directly onto the reduced-form spillover parameter θ in Eq. (9): θ captures the product of ϕ and the sensitivity of the price target to the discount rate.³

A complementary channel operates through narrative coherence. Analysts strive for internal consistency across scenarios: a bear case that emphasizes macro headwinds pulls

³Some DCF implementations weight scenarios explicitly: $P^{\text{reported}} = w_{\text{bear}}P^{\text{bear}} + w_{\text{base}}P^{\text{base}} + w_{\text{bull}}P^{\text{bull}}$. If availability inflates w_{bear} or macro narratives compress P^{bear} , the composite target shifts down. This is a reduced-form version of the discount-rate channel.

the base-case narrative toward acknowledging those same headwinds, even when the base-case probability assessment has not changed. This cognitive-consistency pressure shifts the base-case price target downward independently of the discount rate. While harder to test directly, narrative coherence reinforces the discount-rate channel and helps explain why Macro Bear Bias predicts forecast errors even after controlling for β_i .

This logic also clarifies why comparing ex ante macro attention with ex post realized R^2 is informative. Realized R^2 captures the extent to which firm-level returns are actually linked to market-wide movements. If analysts are correctly attributing downside outcomes to macro risk, then scenarios with more ex ante macro attention should be the ones with a larger conditional market-related component ex post. When macro attention in bear narratives remains elevated even though conditional realized R^2 is similar across scenarios, that wedge is direct evidence of Macro Bear Bias.

The microfoundations above also clarify why a linear specification understates the predictive content of Macro Bear Bias. The availability heuristic generates a macro-attention gap that is roughly state-independent: as documented in Section 2.4, bear-case macro attention is essentially flat across market environments. But the *consequences* of this flat gap are state-dependent. In strong markets, the same macro-heavy bear narrative is more out of line with realized conditions, so the resulting base-case forecast is too pessimistic by a larger margin, generating larger forecast errors. In weak markets, the macro-heavy template is closer to correct, and the distortion is smaller. This asymmetry implies that a linear specification—which assumes the same marginal effect of MacroGap regardless of the market state—compresses genuine predictive content into a single average slope that understates the true signal in favorable environments and overstates it in adverse ones. It is this state-dependent wedge that the nonlinear specification in Section 3.6 is designed to capture, and it explains why out-of-sample R^2 improves dramatically once the model is allowed to interact narrative attention measures with market conditions.

3.4 Forecast Error Prediction

The key implication of the framework is that Macro Bear Bias should predict subsequent errors in analysts' base-case forecasts. Let the realized return and the analyst's base-case forecasted return be

$$R_{i,t+1}^{\text{realized}} \equiv \frac{P_{i,t+1}^{\text{realized}}}{P_t}, \quad R_{i,t+1}^{\text{base}} \equiv \frac{P_{i,t+1}^{\text{base}}}{P_t}, \quad (6)$$

and define the base-case forecast error as

$$\text{Forecast Error}_{i,t+1} = R_{i,t+1}^{\text{realized}} - R_{i,t+1}^{\text{base}}. \quad (7)$$

Under this definition, a positive forecast error indicates that the analyst's base-case forecast is too pessimistic relative to the subsequently realized return.

Absent bias, the analyst's base-case forecast would reflect expected market conditions:

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t, \quad (8)$$

where $\mu_t = E_t[M_{t+1}]$ is the analyst's expectation of the future market excess return. Macro Bear Bias enters when excessive macro emphasis in the bear narrative spills over into the base-case forecast. We capture this spillover through a reduced-form specification: the base-case forecast is shifted downward in proportion to both the firm's systematic exposure β_i —because the bias operates through the macro channel—and the size of the bear–base attention gap. The parameter $\theta > 0$ is a reduced-form coefficient; the testable implication in Eq. (11) holds regardless of the specific functional form.

$$R_{i,t+1}^{\text{base}} = \alpha_{i,t} + \beta_i \mu_t - \theta \beta_i \text{MacroGap}_{i,t}, \quad \theta > 0. \quad (9)$$

This specification formalizes the idea that stronger Macro Bear Bias makes the analyst's central forecast too low. The distortion operates through the macro channel: when analysts

over-attribute downside outcomes to aggregate macro forces, they embed too much downside macro risk into the base case.

Substituting (9) into (7) yields

$$\text{Forecast Error}_{i,t+1} = \beta_i(M_{t+1} - \mu_t) + \theta\beta_i \text{MacroGap}_{i,t} + \varepsilon_{i,t+1}. \quad (10)$$

This equation generates a sharp empirical prediction. Holding fixed the realized market shock and other determinants of firm outcomes, a larger bear-base macro-attention gap should predict a larger, more positive forecast error:

$$\frac{\partial \text{Forecast Error}_{i,t+1}}{\partial \text{MacroGap}_{i,t}} > 0. \quad (11)$$

The intuition is straightforward. A larger $\text{MacroGap}_{i,t}$ indicates that the analyst places unusually strong macro emphasis on the downside scenario relative to the base case. If this reflects Macro Bear Bias rather than a rational assessment of conditional macro risk, then the analyst sets the base-case forecast too pessimistically. When outcomes are later realized, the forecast tends to be too low, producing a positive forecast error.

We now test this prediction empirically. Equation (11) implies that a larger bear-base macro-attention gap should produce a more positive subsequent forecast error, holding fixed the realized market shock. We examine this prediction in two steps. Section 3.5 uses a linear panel regression; Section 3.6 uses a flexible nonlinear specification that allows the effect to vary across market states.

These predictions lead to the following empirical specification:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \text{MacroGap}_{i,t} + b_2 \beta_i + b_3 \text{MacroGap}_{i,t} \times \beta_i + \Gamma' X_{i,t} + u_{i,t+1},$$

where $X_{i,t}$ includes standard firm- and report-level controls. Hypothesis 1 predicts $b_1 > 0$, and Hypothesis 2 predicts $b_3 > 0$.

3.5 Forecast Error Predictability: Empirical Evidence

Guided by the framework in Section 3.4, our central empirical prediction is that stronger Macro Bear Bias should be associated with more positive subsequent base-case forecast errors. Intuitively, when analysts place excessive macro emphasis on downside scenarios, they embed too much downside macro risk into the central forecast. As a result, the base-case forecast becomes too pessimistic relative to subsequently realized returns.

To test this prediction, we estimate the following specification:

$$\text{Forecast Error}_{i,t+1} = a + b_1 \text{MacroGap}_{i,t} + b_2 M_{i,t} + \Gamma' X_{i,t} + \gamma_i + u_{i,t+1}, \quad (12)$$

where $\text{MacroGap}_{i,t}$ is the bear–base gap in macro attention, $M_{i,t}$ is the past 252-day market excess return prior to report release, $X_{i,t}$ is a vector of firm-level controls, and γ_i denotes firm fixed effects. In the baseline specification, $X_{i,t}$ contains only the stock’s own past 252-day return before report release. We include $M_{i,t}$ to control for the market state prevailing when the analyst issues the report. The framework predicts $b_1 > 0$: firms with stronger Macro Bear Bias should exhibit larger positive subsequent forecast errors.

For the empirical analysis, we go beyond the bear–base macro-attention gap and also examine the bull–base macro-attention gap, as well as the corresponding gaps in firm-specific attention as placebo tests. This design helps distinguish Macro Bear Bias from more general differences in narrative intensity across scenarios. If forecast errors are specifically driven by Macro Bear Bias, then the relevant predictor should be the bear–base gap in macro attention, rather than scenario differences in upside narratives or in firm-level risk attention.

The regression results are reported in Table 3. Across columns (1), (5), and (6), the coefficient on the bear–base macro-attention gap is positive and statistically significant, with t -statistics of 2.37, 2.07, and 3.25, respectively. These results indicate that firms for which analysts place relatively greater macro emphasis in the bear case than in the base case subsequently exhibit larger base-case forecast errors. Under our definition of forecast error, this means that analysts set base-case forecasts that are too low for these firms. The evidence

is therefore consistent with Macro Bear Bias distorting analysts' base-case expectations in a systematically pessimistic direction.

By contrast, the bull-base macro-attention gap is not statistically significant, which is consistent with the view that the bias is specific to downside macro narratives rather than to scenario differentiation more broadly. The placebo tests based on firm-specific attention provide weaker and less consistent evidence. Taken together, the results suggest that the predictive content comes primarily from the downside macro component of analyst narratives, in line with the mechanism of Macro Bear Bias.

Table 3 Forecast Error Regressions

Note: This table reports empirical results for forecast error prediction regression (12). Columns (1)–(4) include each gap individually. Column (5) includes the two macro gaps jointly. Column (6) includes all four narrative gap variables simultaneously. Firm fixed effects are included in all specifications. Standard errors are clustered by firm and month. t-statistics are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1)	(2)	(3)	(4)	(5)	(6)
Attention Gaps						
Macro Bear – Base	0.0012** (2.37)				0.0011** (2.07)	0.0023*** (3.25)
Macro Bull – Base		0.0009 (1.03)			0.0005 (0.53)	0.0002 (0.23)
Firm Bear – Base			0.0006 (1.10)			0.0018** (2.28)
Firm Bull – Base				-0.0002 (-0.25)		-0.0004 (-0.57)
Control Variables						
Market Excess Return	-0.1467 (-1.56)	-0.1477 (-1.57)	-0.1474 (-1.57)	-0.1473 (-1.57)	-0.1469 (-1.56)	-0.1470 (-1.57)
Firm Excess Return	-2.3962 (-0.15)	-2.3370 (-0.14)	-2.1720 (-0.13)	-2.3007 (-0.14)	-2.4154 (-0.15)	-2.1880 (-0.13)
Observations	43,313	43,313	43,313	43,313	43,313	43,313
R^2	0.0004	0.0003	0.0003	0.0003	0.0004	0.0006

3.6 Nonlinear Forecast Error Prediction

Table 3 shows that the linear specifications have very limited explanatory power: the in-sample R^2 ranges only from 0.04% to 0.06%, even though the bear–base macro-attention gap significantly predicts subsequent forecast error. This pattern suggests that the predictive content of Macro Bear Bias is not well captured by a simple additive linear specification. While the bear–base macro-attention gap is informative, its effect appears to be state-dependent and therefore difficult to summarize with a single linear slope coefficient.

This interpretation follows naturally from the conceptual framework. Macro Bear Bias is a failure to condition properly on the prevailing market environment. The same bear–base macro-attention gap may therefore imply different forecast distortions depending on whether aggregate conditions are weak or strong. In weak market states, a given amount of downside macro emphasis may map into one degree of pessimism in the base-case forecast; in stronger market states, the same narrative gap may imply a different degree of distortion. More generally, the effect of Macro Bear Bias is likely governed by nonlinear interactions between narrative-based attention measures and market conditions, rather than by their separate linear effects.

To accommodate this nonlinearity, we follow [Kelly, Malamud, and Zhou \(2024\)](#) and employ the random Fourier features (RFF) approach of [Rahimi and Recht \(2007, 2008\)](#). Our raw predictors consist of the macro bear–base gap, macro bull–base gap, firm bear–base gap, firm bull–base gap, and the previous 252-day market excess return prior to report release. We exclude lagged firm-level excess returns so that firm-specific information enters only through the attention measures extracted from analyst reports.

Let G_t denote the 5×1 predictor vector for a given firm-date observation. The RFF procedure maps G_t into nonlinear features,

$$S_{i,t} = [\sin(\gamma\omega'_i G_t), \cos(\gamma\omega'_i G_t)]', \quad \omega_i \sim i.i.d.N(0, I), \quad (13)$$

where each ω_i defines a random projection of the original predictors, which is then trans-

formed through sine and cosine functions. This transformation generates a rich set of nonlinear combinations of attention gaps and market states. As emphasized by Kelly, Malamud, and Zhou (2024), RFF can be viewed as a wide two-layer neural network with random first-layer weights and estimated second-layer coefficients. Its main advantage in our setting is that it introduces flexible nonlinearity while remaining computationally tractable and straightforward to evaluate out of sample.

Following Kelly, Malamud, and Zhou (2024), we use a rolling training window of $T = 12$ and consider models with up to $P = 12,000$ random Fourier features. We set the bandwidth parameter to $\gamma = 2$. For each firm, we generate 12,000 RFFs, estimate recursive out-of-sample forecasts over a grid of model sizes $P \in \{2, \dots, 12,000\}$ and ridge penalties $\log_{10}(z) \in \{-3, \dots, 3\}$, and compute the resulting out-of-sample R^2 . Each firm is forecasted separately using only its own attention-gap variables and the aggregate market state, and the overall out-of-sample R^2 is then computed from the pooled set of firm-level predictions.

Figure A.3 reports the out-of-sample R^2 across model complexity levels. Consistent with Kelly, Malamud, and Zhou (2024), predictive performance initially declines as complexity increases, reaches its trough around $c = 1$, and then improves as complexity rises further. Figure 8 provides a zoomed-in view of the out-of-sample R^2 over the range from -0.05 to 0.05 . The out-of-sample R^2 peaks at approximately 4% when $c = 1000$. Relative to the near-zero in-sample R^2 of the linear model, this improvement is substantial. Overall, the results point to a strongly nonlinear predictive relation: the return implications of Macro Bear Bias are economically meaningful and are most effectively captured by a flexible nonlinear specification that allows the effect of downside macro narratives to vary across market states.

4 Mechanisms

In this section, we discuss two mechanisms that can help explain Macro Bear Bias. The first mechanism is cognitive and arises from the availability heuristic together with anchoring. The second is strategic and arises from career concerns, which can generate a cover-your-

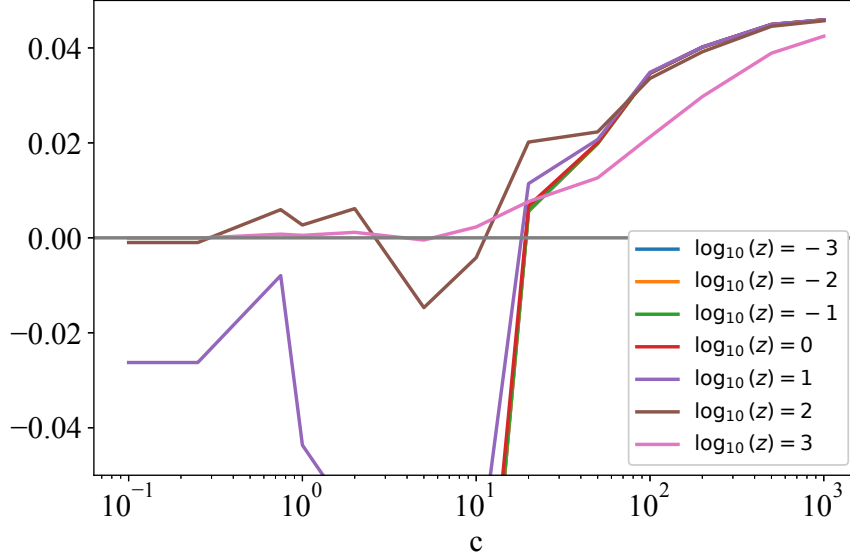


Figure 8 Out-of-Sample Forecast Error Prediction.

Note: This figure reports the out-of-sample R^2 from RFF models used to predict one-year-ahead analyst forecast errors. The training window is $T = 12$ reports and the number of RFFs P ranges from 3 to 12,000 with $\gamma = 2$. The horizontal axis is the complexity level $c = P/T$.

asset (CYA) heuristic in downside communication. Both mechanisms imply that analysts may assign excessive macro emphasis to bear scenarios even when the conditional importance of macro risk is not especially large.

Importantly, these mechanisms are not mutually exclusive. The availability heuristic and anchoring describe how analysts may form distorted downside narratives, while career concerns describe why such narratives may persist even when analysts understand that they are only imperfectly aligned with the conditional environment.

4.1 Availability Heuristic and Anchoring

One natural explanation for Macro Bear Bias is overgeneralization driven by the availability heuristic (Tversky and Kahneman, 1973). When analysts construct downside scenarios, they may not fully condition on the specific market environment associated with the firm and the report at hand. Instead, the bear-narrative template may be anchored to the stock of vivid macro catastrophes in collective analyst memory. The 2008 Global Financial Crisis and the 2020 COVID crash are especially salient exemplars of severe equity downturns: both

were swift, extreme, and overwhelmingly systematic. Because such episodes are vivid and cognitively accessible, analysts may rely on them when constructing downside scenarios, making macro-driven collapses the default bear-case template.

Under this interpretation, analysts do not necessarily make a formal mistake about the sign of macro risk. Rather, they substitute salient downside exemplars for a properly state-contingent assessment of macro relevance. In a rational benchmark, the amount of macro emphasis assigned to a bear narrative should depend on the expected conditional importance of aggregate forces given the prevailing market state. Under the availability heuristic, however, analysts may rely on a simpler rule of thumb: bad outcomes tend to coincide with bad macro environments, so bear cases should be explained in macro terms. This shortcut can generate a persistently positive bear-base macro-attention gap even when the realized macro component of outcomes is similar across scenarios conditional on the same market state.

A key reason this tendency may be asymmetric is that idiosyncratic disasters do not provide an equally available template. Firm-specific collapses are diffuse, case-specific, and narratively incoherent across analysts. They therefore do not supply a shared downside script in the same way that major macro crises do. Macro-driven downturns, by contrast, are collectively observed, widely discussed, and easily redeployed across firms and sectors. As a result, they are more likely to serve as a common interpretive frame for bear narratives.

Anchoring reinforces this mechanism. Once analysts begin to formulate the bear case around an initial macro frame—for example, recession, tightening financial conditions, weak consumer demand, or broad multiple compression—they may insufficiently adjust away from that frame when incorporating firm-specific information. In this sense, the salient crisis episode serves as an anchor, and subsequent reasoning proceeds through partial adjustment around that anchor rather than through a fresh assessment of the firm-specific and state-contingent drivers of downside risk. The resulting narrative then remains excessively macro-heavy.

The viral character of macro narratives further strengthens this anchoring mechanism. As

emphasized by Shiller (2019), recession narratives spread through the investment community and can persist long after fundamentals normalize. This perspective helps explain why Macro Bear Bias may outlive the crisis episodes that originally made macro narratives salient. Consistent with this account, Figure 3 shows that macro bear attention rises significantly around the COVID episode and remains elevated in the subsequent recovery, while the realized R^2 gap narrows considerably once markets stabilize (Figure 7). This divergence—the slow decay of the narrative gap relative to the faster normalization of the empirical benchmark—is the temporal signature of crisis anchoring.

This mechanism also yields a natural cross-sectional implication. If analysts draw on the same pool of recent macro crisis templates, then analysts covering firms in the same industry should exhibit correlated macro emphasis in bear narratives. In other words, availability-based anchoring predicts a form of narrative herding: analysts facing similar sectoral information environments may converge on the same macro-heavy downside framing, even when the underlying firm-specific downside risks differ.

More broadly, the availability-and-anchoring mechanism implies that Macro Bear Bias is fundamentally a failure of conditional attribution. Analysts are not merely pessimistic; they are disproportionately mapping downside scenarios into macro explanations because those explanations are easy to retrieve, widely shared, and difficult to revise once activated.

4.2 Career Concerns and the Cover-Your-Asset Heuristic

A second mechanism is strategic rather than purely cognitive. Analysts may understand that firm-specific forces are often important in downside scenarios, yet still choose to emphasize macro risks because doing so is safer from a career perspective. This channel can be understood as a form of *cover-your-asset* (CYA) heuristic: when discussing adverse outcomes, it is professionally less risky to attribute them to broad external conditions than to firm-specific judgments that may later be proven wrong.

Career concerns provide a natural rationale for why analysts may maintain macro-heavy bear narratives even when a firm’s downside risk is predominantly idiosyncratic. Sell-side

analysts face asymmetric reputational costs: idiosyncratic errors are often more visible and more damaging than consensus-correlated errors (Hong and Kubik, 2003). This asymmetry creates incentives to anchor downside narratives to shared macro risks rather than to sharper firm-level claims.

A macro-driven bear narrative offers at least three advantages in this reputational calculus. First, it is difficult to falsify in real time: if the downside case does not materialize, the analyst can attribute the miss to an unexpectedly resilient macro environment rather than to a mistaken firm-specific judgment. Second, it is relationally neutral: macro claims do not directly implicate management quality or firm-specific execution, which reduces friction in coverage relationships. Third, it is consensus-aligned: anchoring to widely discussed macro risks provides cover from ex post criticism because the analyst’s reasoning is shared with the broader market narrative. A firm-specific bear narrative, by contrast, is more verifiable and often implicitly accusatory.

This logic is especially strong in downside communication. Positive or upside scenarios may reward originality and differentiation, but downside scenarios expose analysts to greater scrutiny because negative calls can be viewed as more consequential, more controversial, or more damaging to issuer relationships and professional standing. In such settings, emphasizing macro headwinds can function as a form of reputational insurance: it frames pessimism as a response to external conditions rather than as a controversial call about the firm itself.

Under this view, Macro Bear Bias reflects not only a belief distortion but also an incentive-compatible communication strategy. Analysts may find it optimal to overweight macro risk in bear narratives because macro-based downside stories are easier to defend to investors, employers, and management teams. This mechanism also helps explain why the bias is asymmetric. The reputational benefit of leaning on macro explanations is likely stronger in downside scenarios than in base or bull scenarios, which naturally generates a larger bear–base macro-attention gap.

The CYA mechanism also yields testable cross-sectional predictions. Analysts at brokerages with active investment-banking relationships should exhibit a larger narrative-realization

gap for covered firms, since the cost of antagonizing management is higher. Junior analysts, who face greater career risk from differentiated misses, should display stronger CYA behavior than more experienced peers. Conversely, analysts known for bold or contrarian calls should exhibit a smaller narrative-realization gap because they are less reliant on consensus-aligned macro framing for reputational protection.

The career-concerns channel also provides a natural link to the asset-pricing results. If analysts strategically frame downside risk in macro terms and allow that framing to pull down the base-case forecast, then the resulting bias should generate predictable forecast errors. Because the pessimistic distortion is systematic rather than purely idiosyncratic, it can also create return predictability when markets do not fully undo the biased component of analyst expectations at the time reports are issued.

4.3 Discussion

The two mechanisms operate through different channels but lead to the same qualitative prediction. Under the availability heuristic and anchoring, analysts overweight macro risk in bear narratives because salient macro downturns are cognitively accessible and serve as anchors for downside reasoning. Under career concerns and the CYA heuristic, analysts overweight macro risk because macro-based downside stories are reputationally safer and easier to defend. In both cases, analysts place excessive macro emphasis on downside scenarios relative to what is justified by the conditional environment.

These mechanisms are also complementary. Availability anchoring explains why macro narratives are the template analysts reach for; career concerns explain why they are reluctant to deviate from that template. The two channels therefore reinforce one another. Salient crises such as 2008 and 2020 make macro narratives especially accessible, while reputational incentives make analysts more willing to retain those macro-heavy narratives than to replace them with sharper firm-specific downside assessments. Together, these mechanisms imply that Macro Bear Bias should be strongest during and after crisis episodes—when macro

narratives are most available—and among analysts facing stronger career or reputation concerns.

Taken together, the availability and career-concerns channels provide a coherent interpretation of why analysts systematically portray bear scenarios as disproportionately macro-driven and why that distortion predicts both subsequent forecast errors and cross-sectional stock returns.

5 Conclusion

We document a systematic pattern in how analysts narrate downside risk. In Morgan Stanley Risk-Reward reports spanning 2007 to 2025, bear-case narratives assign roughly six percentage points more attention to macroeconomic factors than base- or bull-case narratives, a gap that is stable across market regimes, crisis episodes, and expansions alike. We formalize this as a macro bear bias—the failure to condition bear-case narratives on the prevailing market state, over-generalizing the genuine downside comovement documented by [Ang and Chen \(2002\)](#) into an unconditional template. The bias is a miscalibration of source, not magnitude: the bear–base macro-attention gap does not co-vary with realized CAPM R^2 conditional on market state, even as the realized R^2 gap reverses sign from positive in down markets to negative in up markets. The misattribution has measurable downstream consequences. The bear–base attention gap positively predicts subsequent analyst forecast errors, with t -statistics above 3 in the most saturated specification, and is absent for the bull–base gap, confirming that downside overweighting is the operative channel.

Two mechanisms plausibly account for the pattern. The availability heuristic ([Tversky and Kahneman, 1973](#)) predicts that vivid macro catastrophes—the 2008 financial crisis, the 2020 pandemic shock—anchor analysts’ default bear-case template and that this anchor decays slowly, generating macro-heavy narratives well into the subsequent recovery. Career concerns provide a complementary incentive channel ([Holmström, 1999](#); [Scharfstein and Stein, 1990](#)): macro-attributed downside is harder to falsify in real time and less reputation-

ally costly than firm-specific attribution, making macro framing a strategically safe default. Both channels predict that the narrative gap widens during and immediately after macro crises.

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INTERNET APPENDIX

A LLM Prompt

You are a financial analyst trained in equity valuation and empirical asset pricing. You will analyze the Bull, Base, and Bear case texts from a Morgan Stanley equity research report. Your task is to extract reasoning, classify it as firm-level, industry-level, or macro-level, identify the economic channel, and quantify attention shares across cases. Write in a concise, research-style tone used in equity research reports.

Return your full answer ****strictly in JSON format****, with one JSON object per report (suitable for .jsonl storage). Each JSON object must include keys: "filename", "bull_case", "base_case", "bear_case", and "cross_case_summary". Do not include explanations outside the JSON.

Tasks:

1. Price Targets: List the bull case, base case, and bear case price targets stated in the Risk Reward section.
2. Reason Extraction: Identify all distinct reasons for the bull, base, and bear cases. Requirements:
 - 2.1 Include any explicitly stated upside risks as part of the bull-case reasons.
 - 2.2 Include any explicitly stated downside risks as part of the bear-case reasons.
 - 2.3 Preserve the original wording as closely as possible (avoid paraphrasing).
 - 2.4 Number each reason sequentially (e.g., 1., 2., 3.).
 - 2.5 Label each reason as one of the following categories: Firm, Industry, Macro, or Mixed. If labeled Mixed, specify which component (Firm, Industry, or Macro) is dominant.
 - 2.6 Assign each reason to a single economic channel that best represents

its underlying mechanism, such as expected cash flow, discount rate, risk aversion or asset demand.

- 3.Attention Quantification: Quantify the analyst's relative attention across Firm, Industry, and Macro news as: (Firm %, Industry %, Macro %), ensuring $\text{Firm \%} + \text{Industry \%} + \text{Macro \%} \leq 100$.
- 4.Attention Evolution Summary: Summarize how the analyst's attention distribution across Firm, Industry, and Macro news (from Step 2.5) evolves across the bull, base, and bear cases.
- 5.Channel Evolution Summary: Summarize how the dominant economic channels (from Step 2.6) evolve across the bull, base, and bear cases.

Output format (strict JSON):

```
{
  "filename": "<report_name>",
  "bull_case": {
    "bull_price_target": "<numerical price target or range>",
    "reasons": [
      {
        "id": 1,
        "text": "<verbatim reason>",
        "category": "Firm | Industry | Macro | Mixed (dominant: Firm/
          Industry/Macro)",
        "economic_channel": "Expected Cash Flow | Discount Rate | Risk
          Aversion | Asset Demand"
      }
    ],
    "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
    "analytical_summary": "[one concise sentence summarizing key focus of
      bull case]"
  },
  "base_case": {
    "base_price_target": "<numerical price target or range>",
```

```

"reasons": [
  {
    "id": 1,
    "text": "<verbatim reason>",
    "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
    "economic_channel": "Expected Cash Flow | Discount Rate | Risk
      Aversion | Asset Demand"
  }
],
"attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
"analytical_summary": "[one concise sentence summarizing key focus of
  base case]"
},

"bear_case": {
  "bear_price_target": "<numerical price target or range>",
  "reasons": [
    {
      "id": 1,
      "text": "<verbatim reason>",
      "category": "Firm | Industry | Macro | Mixed (dominant: ...)",
      "economic_channel": "Expected Cash Flow | Discount Rate | Risk
        Aversion | Asset Demand"
    }
  ],
  "attention_distribution": { "Firm": X, "Industry": Y, "Macro": Z },
  "analytical_summary": "[one concise sentence summarizing key focus of
    bear case]"
},

"cross_case_summary": {
  "attention_evolution": "[summary of how Firm/Industry/Macro attention
    shifts across bull -> base -> bear]",

```

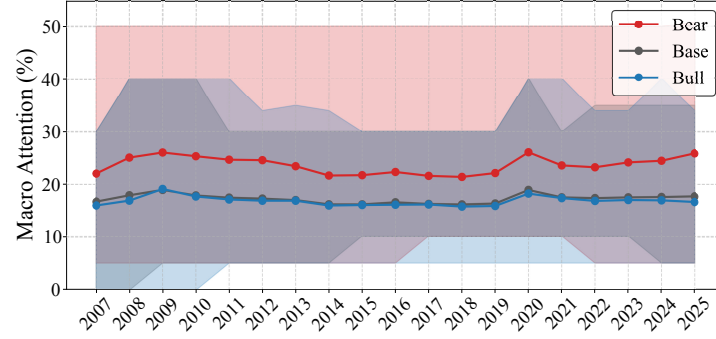
```
    "channel_evolution": "[summary of how dominant economic channels shift  
        across bull -> base -> bear]"  
  }  
}
```

Rules:

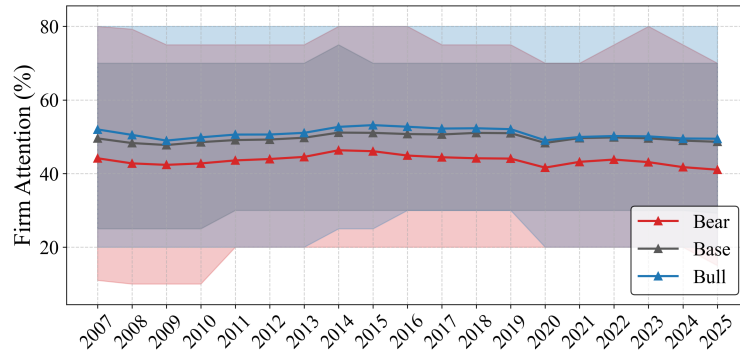
- All percentages must be integers (no decimals).
- Maintain an analytical and academic tone throughout.
- Do NOT include any text outside the JSON structure.
- Ensure structural consistency and valid JSON formatting.

B Additional Exhibits

Panel A: Macro Attention



Panel B: Firm Attention



Panel C: Industry Attention

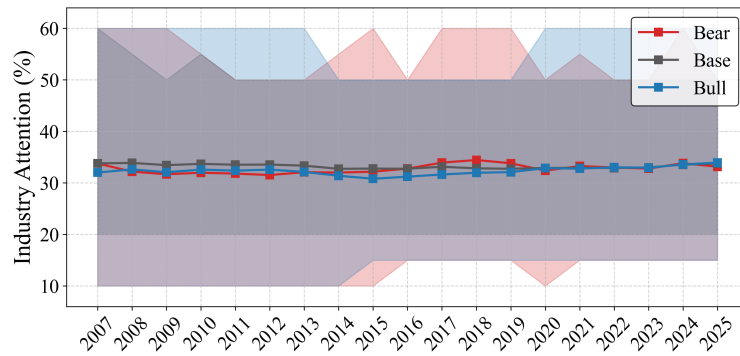


Figure A.1 Time Series of Attention to Firm, Industry, and Macro News.

Note: Time series of annual average attention shares to firm-, industry-, and macro-level news. The solid lines represent yearly means. The shaded areas correspond to the empirical 5th to 95th percentile range of the cross-sectional distribution in each year.

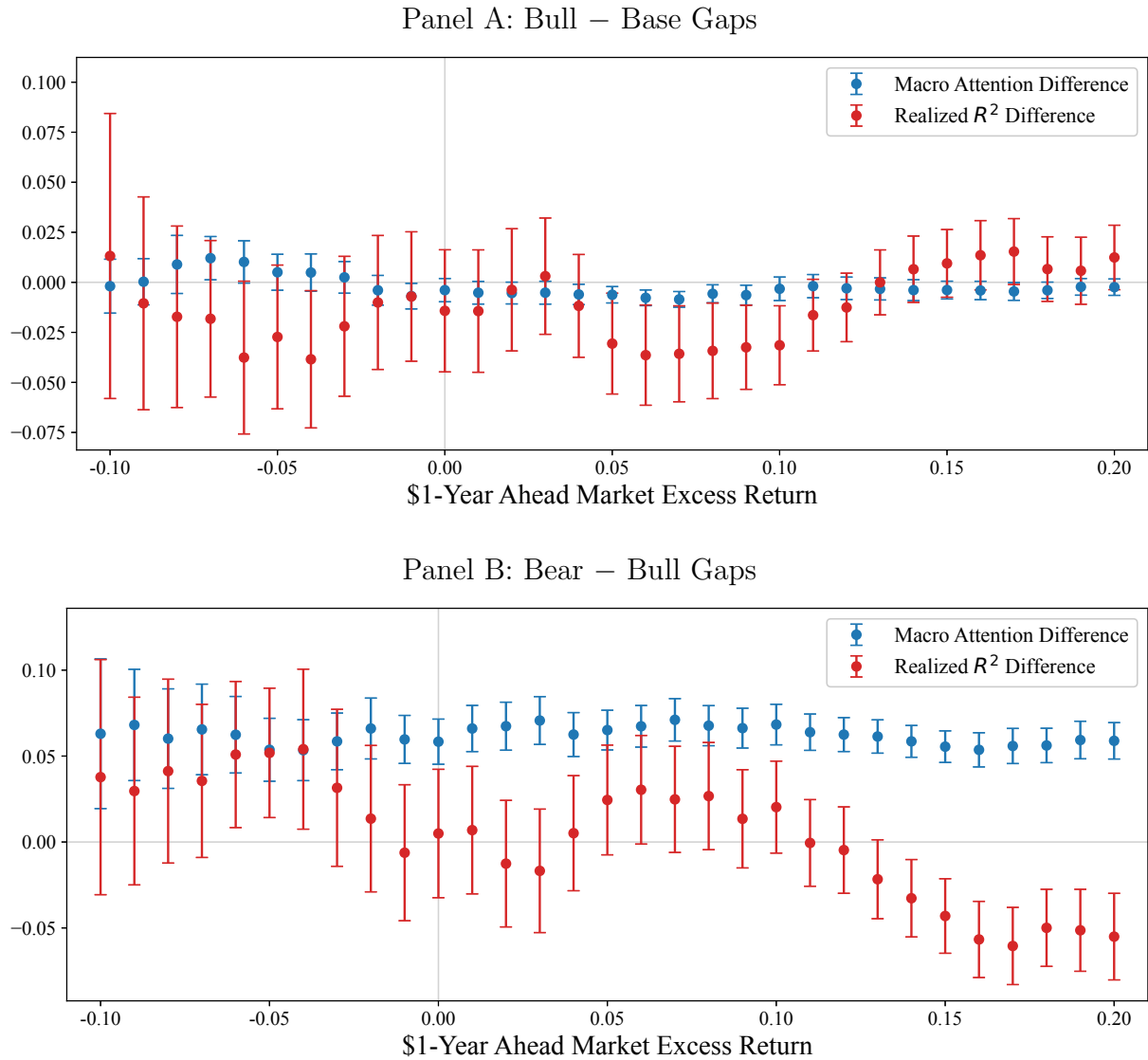


Figure A.2 Gaps in Macro Attention and Realized CAPM R^2 Conditional on Market States.

Note: This figure plots firm-level bull–base and bear–bull differences in macro attention and realized CAPM R^2 across subsamples defined by 10% bins of forward 252-day market excess returns following the analyst report release. Each point includes reports with $x - 5\% \leq (R_m - R_f) \leq x + 5\%$ and applies the same-firms rule within each bin, requiring firms to have realized-bear, realized-base, and realized-bull reports. Macro attention is measured as the firm-level average macro attention in the bull, base, and bear cases. Realized CAPM R^2 is computed as the firm-level average R^2 over the 252 trading days following the report release for reports ex post classified as realized bull, realized base, and realized bear cases. The plotted values represent the bull-minus-base and bear-minus-bull differences. Error bars show 95% confidence intervals from constant-only regressions.

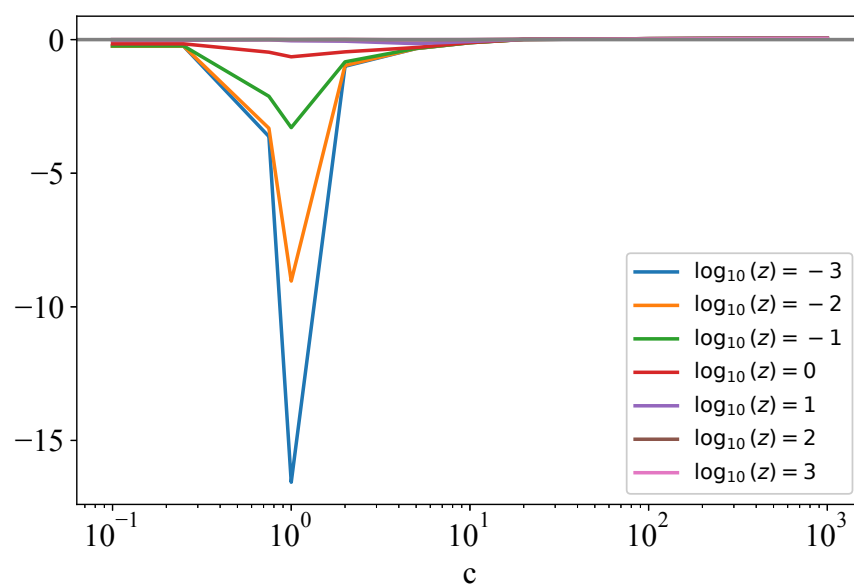


Figure A.3 Out-of-sample Market Timing Performance.

Note: This figure shows the out-of-sample R^2 that use RFFs to predict one-year ahead forecast error. The training window is $T = 12$ reports and RFF count P ranges from 3 to 12,000 with $\gamma = 2$. The horizontal axis is the complexity level $c = P/T$.